DANIEL KARIMI

SOFTWARE DEVELOPMENT ENGINEER

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ABOUT ME

Engineering graduate (Universidad Politécnica de Madrid, 2025/2026) with a strong foundation in **Object-Oriented Design, Data Structures, and Algorithms** using Java, Python, and C++, and deep knowledge of the **full Software Development Life Cycle (SDLC)**. I am passionate about designing and coding the right solutions to broadly defined, complex engineering problems — building systems that are scalable, fault-tolerant, and easy to operate.

At CSIC, I architected and delivered complex, end-to-end software systems from scratch to production-ready state. I built scalable, modular pipelines using clean software design patterns, contributed actively to architecture and reliability decisions in a distributed environment, and applied professional development practices including **Git, CI/CD, rigorous testing, and technical documentation**. My code directly contributed to novel scientific results that are pending publication.

I am excited to join **Amazon Business** as an SDE, bringing my ownership-driven mindset and engineering discipline to build distributed back-end services that scale to millions of B2B customers across multiple countries and regions.

TECHNICAL STACK

💻 CORE SOFTWARE ENGINEERING

Java • Python • C / C++ •
Object-Oriented Design •
Data Structures • Algorithm Design

🏗️ ARCHITECTURE & SYSTEMS

Distributed Systems •
Scalable Architecture •
Fault-Tolerant Design •
Microservices Concepts •
Design Patterns • System Reliability

☁️ CLOUD & BACKEND

AWS (Familiarity) • RESTful APIs •
Docker • Distributed Storage •
Cloud-Native Design •
Indexing & Query Systems

🛠️ DEV PRACTICES & SDLC

Git / GitHub • CI/CD Pipelines •
Agile / Scrum • Code Reviews •
Unit Testing & QA •
Operational Excellence

🐧 SYSTEMS & OS

Linux / Unix OS • Networking (TCP/IP) •
Bash Scripting •
Debugging & Troubleshooting •
Performance Optimization

🤖 AI / ML EXPOSURE

PyTorch / TensorFlow •
ML Pipeline Engineering •
Data Processing at Scale •
Prediction Systems • Automation

Languages: Spanish (Native) • English (B2 – Aptis ESOL, British Council)

PROFESSIONAL EXPERIENCE

Software & AI Engineer Intern | CSIC – Institute of Physical and Information Technologies (ITEFI)

Sep 2025 – Feb 2026

- Designed and built **scalable, fault-tolerant software systems** in Python from scratch, contributing concrete architectural decisions around modularity, data flow, and component reliability across a distributed research environment.
- Engineered end-to-end data processing pipelines with a focus on **high performance and scalability**, reducing execution times from hours to minutes for large multi-dataset workloads.
- Applied professional **SDLC practices** across the full project lifecycle: technical requirements analysis, architecture definition, iterative development, code review, testing, and documentation using Git for source control.
- Contributed to **architecture decisions** for storage and processing components, applying software design patterns (OOP) to ensure clean, maintainable, and easily extensible codebases.
- Collaborated in an agile, cross-functional team of scientists and engineers, communicating technical solutions clearly to both technical and non-technical stakeholders through written reports and presentations.

Python | Distributed Architecture | OOP | Scalable Systems | CI/CD | Git | Agile

EDUCATION

Telecommunications Systems Engineering (BSc) | **Universidad Politécnica de Madrid (ETSIT-UPM)** *Sep 2020 – Jul 2025*

- **Programming, Algorithms & OOP:** Rigorous academic training in Data Structures, algorithm complexity, and Object-Oriented Design using **Java, Python, and C/C++**. Core foundation for architecting well-structured software.
- **Distributed Systems & Networking:** Deep study of TCP/IP protocols, distributed operating systems, scalable system design, and network reliability — directly applicable to building distributed storage and query systems.
- **Mathematical & Analytical Foundations:** Advanced mathematics (Linear Algebra, Statistics, Calculus) and DSP providing rigorous problem-solving methodologies for prediction systems and large-scale processing.

Java | C/C++ | Python | Algorithms | Distributed Systems | Networking

Technology Baccalaureate, **IES Isabel la Católica** *2018 – 2020*

Science and Technology track. Honours. Strong mathematical and analytical foundation.

CERTIFICATIONS

AI and Productivity (Google): Leveraging AI tools for development productivity and automated workflow design. •

Generative AI (MIT): Design and implementation of GenAI solutions, including scalable data synthesis and intelligent system integration. •

Internet of Things (MIT): Distributed device architectures, sensor data pipelines, and cloud integration at scale. •

Aptis ESOL B2 (British Council): Professionally certified English communication skills — essential for collaborative presentations, workshops, and technical evangelism in international teams.

KEY PROJECTS

CellTracker (Distributed Processing System) ARCHITECTURE DESIGN

PYTHON • OOP • DISTRIBUTED DESIGN • ALGORITHMS

End-to-end scalable system pending publication in a peer-reviewed scientific journal.

Designed and coded an end-to-end software system integrating ML-based detection models with algorithmic tracking components (Kalman Filters). Led architecture decisions across storage, processing, and query layers to ensure scalability, fault tolerance, and low operational overhead.

- Applied object-oriented design patterns to build highly modular, reusable components deployable across different environments.
- Delivered a functionally complete and operationally stable system that autonomously processes large datasets, directly contributing to novel scientific results.

MigraCell Accelerator SCALABLE BACKEND

PYTHON • HIGH-PERFORMANCE PIPELINES • ML INTEGRATION

Scalable data processing service that drastically improves performance at high data volumes.

Built a high-performance back-end service integrating Deep Learning models with automated data extraction and indexing pipelines. Focused on cost efficiency and operational stability — reducing processing overhead from hours to minutes while maintaining output quality and reproducibility.

- Translated broadly defined problem requirements into scalable, well-architected technical deliverables.
- Engrained operational excellence across the codebase: automated testing suites, performance metrics monitoring, and clear documentation for every system component.